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Please write clearly in block capitals.

Centre number | | | | | |

Candidate number | | | | |

Surname

Forename(s)

Candidate signature

Computer Science

Paper 2 topics: Computational thinking, algorithms and programming (2.1 to 2.5)

Suggested time: about 30 minutes. There is no timer.

Materials

For this paper you must have:

- a pen
- no calculator is allowed

Instructions

- Use black ink.
- Write your answer to each question in the space provided.
- Answer all the questions.
- Where you're asked to write an algorithm, you can use OCR Exam Reference Language or Python.

Information

- The total mark for this paper is 30.
- The marks for each question are shown in brackets [ ].

For Examiner's Use	
Question	**Mark**
1	
2	
3	
4	
5	
TOTAL	

This is a practice paper written for Dougie in the style of the GCSE exam. It is not an exam board paper, and it is not a paper from any past exam series.

1 **(a)** State what is meant by abstraction.

[1]

(b) State what is meant by decomposition.

[1]

(c) Here is a list of numbers: 3, 8, 12, 15, 21, 27, 30.

Explain why a binary search can be used on this list.

[1]

(d) Show the steps a binary search would take to find the number 21 in the list 3, 8, 12, 15, 21, 27, 30.

[3]

(e) A bubble sort is used to put the list 5, 2, 9, 1 into ascending order. Write the list as it is after the first pass.

[2]

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2 **(a)** Look at this code:

```
total = 0
for i = 1 to 4
   total = total + i * 2
next i
print(total)
```

Give the value of total after each time round the loop, and the value that is printed.

[2]

(b) State the type of iteration used in this code.

[1]

(c) State the most suitable data type for each of these values: 15, 3.5, "Yes", True.

[2]

(d) What is the result of 17 MOD 5?

[1]

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(e) Write an algorithm that asks the user to enter a password. It should keep asking until they enter letmein, then print Access granted. You can use OCR Exam Reference Language or Python.

[3]

3 **(a)** A program asks for a student's age. The age must be from 11 to 18. Name a validation check that could be used, and describe what it would check.

[2]

(b) Give one example of each type of test data for the age: normal, boundary and erroneous.

[3]

(c) Give one way a programmer can make their code easier to maintain.

[1]

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4 **(a)** P = A AND NOT B. Write the value of P for each row of the table, from top to bottom.

A	B	P
0	0	?
0	1	?
1	0	?
1	1	?

[2]

(b) Name the logic gate that outputs 1 only when both of its inputs are 1.

[1]

5 **(a)** Give two differences between a compiler and an interpreter.

[2]

(b) Name one feature of an IDE (integrated development environment) and say how it helps a programmer.

[2]

END OF QUESTION PAPER

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